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United States Patent Application Publication

20250282454

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Peterson; Jeffrey

Braking System for Gyroscopic Boat Roll Stabilizer

Abstract

A gyroscopic boat roll stabilizer for a boat comprises a gimbal, having a gimbal axis, an enclosure mounted to the gimbal and configured to precess about a gimbal axis, a flywheel assembly rotatably mounted inside the enclosure for generating a torque that is applied to counter a rolling motion of the boat, and a braking system for controlling precession of the enclosure.

Inventors: Peterson; Jeffrey (Mooresville, NC)

Applicant: WaveTamer LLC (Greenville, NC)

Family ID: 1000008629199

Appl. No.: 19/219125

Filed: May 27, 2025

Related U.S. Application Data

parent US continuation 18624362 20240402 parent-grant-document US 12312050 child US 19219125

parent US continuation 18385651 20231031 parent-grant-document US 11964735 child US 18624362

parent US continuation 18127870 20230329 parent-grant-document US 11820469 child US 18385651

parent US continuation 17826860 20220527 parent-grant-document US 11628914 child US 18127870

parent US continuation 16832510 20200327 parent-grant-document US 11440629 child US 17826860

us-provisional-application US 62828845 20190403

Publication Classification

Int. Cl.: B63B39/04 (20060101)

U.S. Cl.:

CPC B63B39/04 (20130101);

Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/624,362, filed 2 Apr. 2024, which is a continuation of U.S. patent application Ser. No. 18/385,651, filed 31 Oct. 2023, now U.S. Pat. No. 11,964,735; which is a continuation of U.S. patent application Ser. No. 18/127,870, filed 29 Mar. 2023, now U.S. Pat. No. 11,820,469; which is a continuation of U.S. patent application Ser. No. 17/826,860, filed 27 May 2022, now U.S. Pat. No. 11,628,914; which is a continuation of U.S. patent application Ser. No. 16/832,510, filed 27 Mar. 2020, now U.S. Pat. No. 11,440,629; and claims benefit of U.S. Provisional Patent Application Ser. No. 62/828,845, filed 3 Apr. 2019; the disclosures of all of which are incorporated by reference herein in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to boat roll stabilizers for reducing the sideways rolling motion of a boat and, more particularly, to an improved braking system for controlled moment gyroscopes.

BACKGROUND

[0003] The sideways rolling motion of a boat can create safety problems for passengers and crew on boats, as well as cause discomfort to passengers not accustomed to the rolling motion of the boat. A number of technologies currently exist to reduce the sideways rolling motion of a ship. One technology currently in use is active fin stabilization. Stabilizer fins are attached to the hull of the ship beneath the waterline and generate lift to reduce the roll of the ship due to wind or waves. In the case of active fin stabilization, the motion of the ship is sensed and the angle of the fin is controlled based on the motion of the ship to generate a force to counteract the roll. Fin stabilization is most commonly used on large ships and are effective when the ship is underway. Fin stabilization technology is not used frequently in smaller boats and is generally not effective when the boat is at rest. Stabilizer fins also add to the drag of the hull and are susceptible to damage.

[0004] Gyroscopic boat stabilization is another technology for roll suppression that is based on the gyroscopic effect. A control moment gyroscope (CMG) is mounted in the boat and generates a torque that can be used to counteract the rolling motion of the boat. The CMG includes a flywheel that spins at a high speed. A controller senses the attitude of the boat and uses the energy stored in the flywheel to “correct” the attitude of the boat by applying a torque to the hull counteracting the rolling motion of the boat. CMGs work not only when a boat is underway, but also when the boat is at rest. CMGs are also less expensive than stabilizer fins, do not add to the drag of the hull, and are not exposed to risk of damage.

[0005] Although CMGs are gaining in popularity, particularly for smaller fishing boats and yachts, this technology has some limitations. CMGs rely on a braking system to control the precession of the flywheel. In prior art CMGs, the design of the braking system constrains the precession of the flywheel to about ± 22 degrees, which may not be sufficient to effectively counter the rolling motion of the boat. Further, the resistance of the braking systems imposes limitations on the rate of precession, which affects the response of the CMG to the wave motion.

SUMMARY

[0006] The present disclosure relates to a gyroscopic boat roll stabilizer configured to be installed in a boat. The boat roll stabilizer comprises a gimbal, having a gimbal axis, an enclosure mounted to the gimbal and configured to precess about a gimbal axis, a flywheel assembly rotatably mounted inside the enclosure for generating a torque that is applied to counter a rolling motion of the boat, and a braking system for controlling precession of the enclosure. The braking system is configured to enable precession in the first and second directions of up to at least 45 degrees.

[0007] According to another aspect of the disclosure, the braking system comprises a first actuator connected between the support frame and the enclosure to resist precession in a first direction about the gimbal axis, and a second actuator connected between the support frame and the enclosure to resist precession in a second direction about the gimbal axis. The first actuator and the second actuator both connect to the enclosure on the same side of a transverse plane including the gimbal axis, and on different sides of a frontal plane including the gimbal axis.

[0008] According to another aspect of the disclosure, each actuator comprises a fluid cylinder and a lockout valve mounted to the fluid cylinder of the actuator. The lockout valve is movable between a locked position preventing precession of the enclosure and an unlocked position. The lockout valves are in fluid communication with the piston side of the fluid cylinders. In one embodiment, a manifold is disposed in a fluid flow path between the first and second actuators. A first port on each lockout valve is in fluid communication with the manifold, while a second port on each lockout valve is in fluid communication with the rod side of the fluid cylinder. A fluid line connects the first port of the lockout valve on the actuator to the manifold.

[0009] According to another aspect of the disclosure, a gyroscopic boat roll stabilizer configured to be installed in a boat includes a gimbal, a flywheel assembly, and a braking system. The gimbal includes a support frame and enclosure configured to precess about a gimbal axis. The flywheel assembly is rotatably mounted inside the enclosure for generating a torque that is applied to counter a rolling motion of the boat. The braking system controls precession of the enclosure, and includes a first actuator and a manifold assembly. The manifold assembly is connected in a fluid flow path to the first actuator, and the manifold assembly includes a main valve. The main valve is connected to the first actuator by a fluid line that allows fluid flow between the main valve and the first actuator. The manifold assembly further comprises an hydraulic accumulator operatively connected to the main valve on a low pressure side of the main valve. In some embodiments, the manifold assembly further comprises a second hydraulic accumulator operatively connected to the main valve on the low pressure side of the main valve, with the first accumulator associated with the first actuator and the second accumulator associated with a second actuator.

[0010] According to another aspect of the disclosure, each of the actuators further comprises a bypass line connecting the second port of the lockout valve and configured to communicate fluid directly between the cylinder side and rod side of the fluid cylinder of the actuator without having to pass through the main valve.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIGS. 1A and 1B illustrate a boat equipped with a boat roll stabilizer.

[0012] FIG. 2 is a perspective view of the boat roll stabilizer showing the gimbal and enclosure.

[0013] FIG. 3 is a cross section of the enclosure for the boat roll stabilizer, showing the flywheel assembly.

[0014] FIG. 4 is an elevation view of a boat roll stabilizer with the enclosure in a neutral or vertical position.

[0015] FIG. 5 is an elevation view of a boat roll stabilizer with the enclosure rotated at a 45° angle.

[0016] FIG. 6 is a schematic block diagram of a torque control system for the boat roll stabilizer.

[0017] FIG. 7 is a perspective view of a braking system for a boat roll stabilizer.

[0018] FIG. 8 is a schematic diagram of a braking system for a boat roll stabilizer.

[0019] FIG. 9 is a perspective view of the manifold for the braking system.

[0020] FIG. 10 is a section view of the manifold for the braking system.

[0021] FIG. 11 is a graph of the roll resistance to a large rolling wave of a boat roll stabilizer. as herein described compared to the roll resistance of a conventional boat roll stabilizer.

[0022] FIG. 12 is a graph of the roll resistance to a large short wave of a boat roll stabilizer. as herein described compared to the roll resistance of a conventional boat roll stabilizer.

DETAILED DESCRIPTION

[0023] Referring now to the drawings, FIGS. 1A and 1B illustrate a CMG 10 mounted in a boat 5 for roll stabilization. Multiple embodiments of the CMG 10 are described. For convenience, similar reference numbers are used in the following description of the embodiments to indicate similar elements in each of the embodiments.

[0024] The main functional elements of the CMG 10 comprise a single-axis gimbal 20, an enclosure 30 mounted to the gimbal 20 for rotation about a gimbal axis, a flywheel assembly 40 mounted by bearings 50 inside the enclosure 30, a motor 60 to rotate the flywheel assembly 40, and a torque control system 100 (FIG. 6) to control precession of the flywheel 42 so that the energy of the flywheel assembly 40 is transferred to the hull of the boat 5 to counteract rolling motions.

[0025] The gimbal 20 comprises a support frame 22 that is configured to be securely mounted in the boat 5. Preferably, the gimbal 20 is mounted along a longitudinal axis (LA) of the boat 5 with the gimbal axis G extending transverse to the longitudinal axis of the boat 5. Conventionally, the gimbal 20 is mounted in the hull of the boat 5, but could be mounted at any location. The enclosure 30 is rotatably mounted to the support frame 22 by spherical roller bearings 28 housed in pillow blocks so as to rotate about the gimbal axis G extending transversely across the boat 5. For this purpose, the enclosure 30 includes two gimbal shafts 32 projecting from diametrically opposed sides of the enclosure 30. The gimbal shafts 32 are received in the spherical bearings 28 to allow the enclosure 30 and flywheel assembly 40 to rotate or precess about the gimbal axis G in the fore and aft directions.

[0026] The enclosure 30 is generally spherical in form and comprises two main housing sections 34 and two cover plates 36. The two main housing section 34 join along a plane that bisects the spherical enclosure 30. The cover plates 36 join the main housing sections along respective planes closer to the “poles” of the spherical enclosure 30. All joints in the enclosure 30 are sealed to maintain a below-ambient pressure within the enclosure 30 to reduce aerodynamic drag on the flywheel assembly 40.

[0027] The flywheel assembly 40 comprises a flywheel 42 and shaft 44 that is mounted for rotation inside the enclosure 30 so that the rotational axis F of the flywheel 42 is perpendicular to the gimbal axis G. Thus, when the boat 5 is level, the rotational axis F of the flywheel shaft 44 will be in the vertical direction, i.e., perpendicular to the deck of the boat. The flywheel 42 and shaft 44 may be formed as a unitary piece, or may comprise two separate components. In one exemplary embodiment, the diameter of the flywheel 42 is approximately 20.5 inches and the flywheel assembly 40 has a total weight of about 614 lbs. The flywheel assembly 40 has a moment of inertia of about 32,273 lb in.sup.2. When rotated at a rate of 9000 rpm, the angular momentum of the flywheel assembly 40 is about 211,225 lbm ft.sup.2/s.

[0028] The flywheel assembly 40 is supported by upper and lower bearing assemblies inside the enclosure 30. Each bearing assembly comprises a bearing 50 mounted within a bearing block 58. Each bearing 50 comprises an inner race 52 that contacts and rotates with the flywheel shaft 44, an outer race 54 that is fixed inside the bearing block 58, and a ball 56 disposed between the inner and outer races 52, 54. The bearing blocks 58 are secured to the interior of the enclosure. Seals (not shown) are disposed on the top and bottom of the bearings 50 to contain lubricant in the bearings 50. Each end of the flywheel shaft 44 includes a cavity with a heat sink inserted therein to dissipate

heat as described in U.S. Provisional Application No. 62/768,356 filed Nov. 18, 2018, which is incorporated herein in its entirety by reference.

[0029] The motor **60** rotates the flywheel assembly **40** at a high rate of speed (e.g., 9000 rpm). Although the motor **60** is shown mounted inside the enclosure **30**, it is also possible to mount the motor **60** on the exterior of the enclosure **30**. In one embodiment, the motor **60** comprises operates on 230 volt single phase AC power and is able to accelerate a flywheel assembly with a moment of inertia of about 32,273 lb in.^{sup.2} flywheel from rest to a rotational speed of 9000 rpm preferably in about 30 minutes or less for an average acceleration of about 5 rpm/s, and more preferably in about 20 minutes or less for an average acceleration of about 7.75 rpm/s, and even more preferably in about 10 minutes or less for an average acceleration of about 15 rpm/s (or 1.57 radians/s.^{sup.2}). [0030] The torque control system **100**, shown in FIG. **6**, controls the precession of the flywheel **42** about the gimbal axis. The rolling motion of a boat **5** caused by wave action can be characterized by a roll angle and roll rate. The rolling motion causes the flywheel **42** to precess about the gimbal axis. Sensors **104**, **106** measure the roll angle and roll rate respectively, which are fed to a controller **102**. The controller **102** generates control signals to control an active braking system **110** or other torque applying device that controls the precession of the flywheel **42**. By controlling the precession, the flywheel assembly **40** generates a torque to counteract the rolling motion. This torque is transferred through the gimbal **20** to the boat **5** to dampen the roll of the boat **5**.

[0031] In conventional braking systems **110** for CMGs **10**, the design of the braking system constrains the precession of the flywheel to about ± 22 degrees, which may not be sufficient to effectively counter the rolling motion of the boat. Further, the resistance of the braking systems **110** imposes limitations on the rate of precession, which affects the response of the CMG **10** to wave motion.

[0032] According to one aspect of the disclosure, the braking system **110** is designed to enable the flywheel **42** to precess up to about ± 45 degrees. An exemplary embodiment of the braking system **110** is shown in FIGS. **7** and **8**. FIG. **7** is a reverse perspective view of the braking system **110**. As seen in FIG. **7**, the braking system **110** comprises a pair of actuator assemblies **120** that control the precession of the flywheel **42** about the gimbal axis, and a manifold assembly **150** for transferring fluid between the opposing cylinders **120** when the flywheel precesses. Hydraulic lines connect the fluid cylinders **120** with the manifold assembly **150** to form a closed fluid flow path between the cylinders **120**.

[0033] FIG. **8** is a schematic diagram of the braking system **110**. As seen in FIG. **8**, the actuator assemblies **120** each comprise a fluid cylinder **122** having a housing **124**, piston **126**, and piston rod **128**. The fluid cylinders **122** each include a piston side port and a rod side port. A lockout valve **130** is mounted on the housing **124** of the fluid cylinder **122** and is in fluid communication with the piston side port of each fluid cylinder **122**. The lockout valve **130** is normally open and is closed by actuation of a solenoid **132**. In one embodiment, the housing of the lockout valve **130** is secured to or integrally formed with the housing **124** of the fluid cylinder **122** to form a unitary assembly. Each fluid cylinder **122** is pivotally connected at one end to the support frame **22** and at the other end to the enclosure **30**. The housing **124** of the fluid cylinder **122** pivotally connects to the support frame **22** and the piston rod **128** pivotally connects to the enclosure **30** via a connecting plate **140** (FIG. **7**), although this arrangement could be revised.

[0034] The connecting plate **140**, shown best in FIG. **7**, bolts to the exterior of the enclosure **30** and includes two pivot pins **142** that are rotatably journaled in bushings or bearings (not shown) disposed at the end of respective piston rods **128**. The connecting plate **140** is symmetrical about a frontal plane and the pivot pins **142** are offset from the frontal plane by a distance D. As used herein, the term "frontal plane" refers to a vertical plane (when the enclosure **30** is in a neutral position) that includes the gimbal axis and divides the enclosure **30** into front and back sections. The axes of the pivot pins **142** are parallel to the frontal plane. Three bolts **144** pass through corresponding openings in the connecting plate **140** and thread into threaded holes (not shown) in

the enclosure **30** to secure the connecting pale **140** to the enclosure.

[0035] Due to the mechanical arrangement of the braking system **110**, the enclosure **30** is able to precess up to about ± 45 degrees without any mechanical interference compared to about ± 22.5 degrees for prior art designs. The greater degree of precession enables a greater peak roll torque to be achieved.

[0036] Referring back to FIG. **8**, the manifold assembly **150** comprises a main valve **152**, two check valves **154**, two pressure relief valves **156**, and two accumulators **158**. The main valve **152** is mounted to the manifold block **151** as shown in FIG. **9** and has four ports labeled A, B, P and T respectively and is controlled by a solenoid **134**. Ports P and B connect via fluid lines **160A** and **160B** to the normally closed lockout valves **130** in actuator assemblies **120A** and **120B** respectively. Ports A and T connect the main valve **152** to respective accumulators **158**. Fluid flowing into the main valve via port P exits via port A. Fluid flowing into the main valve **152** via port B exits via port T. The check valves **154** control the direction of the fluid flow through the main valve **152** and prevent back flow into ports A and T of the main valve **152**. The pressure relief valves **156** prevent over-pressurization of the fluid in the fluid components of the braking assembly **10** due to thermal expansion of fluid when the enclosure **30** is locked. The main valve **152** and lockout valves **130** are controlled by solenoids **132** and **134** respectively, which are actuated by the controller **102**.

[0037] When the enclosure **30** precesses such that the piston **126** of fluid cylinder **122A** retracts, fluid flows from the piston side of the fluid cylinder **122A** through the lockout valve **130**. A portion of the fluid exiting the fluid cylinder **122** flows through a bypass line **162A** to the rod side of the fluid cylinder **122A**. The rod side of the cylinder is unable to accommodate all of the fluid exiting the piston side of the cylinder due to the volume of the rod **128**. Therefore, a portion of the fluid flows into port P of the main valve **152**. From the main valve **152**, the fluid exits port A, flows through the check valve **154**, which prevents backflow into the manifold **150** and through line **160B** to the lockout valve **130B** on the piston side of the fluid cylinder **122B**. When the enclosure **30** precesses in the opposite direction, the fluid flow reverses.

[0038] The accumulators **158** provide additional capacity in case the fluid expands due to heat, or due to imbalance of the fluid flow. The heating or imbalance of the fluid flow will create a higher pressure in the main valve **152**, which in turn will cause any excess fluid to flow via ports A and T into the accumulators **158**. Because ports A and T are on the low pressure side of the main valve, the cycling of fluid into and out of the accumulators is minimized.

[0039] In some embodiments, a cooling circuit **170** can be provided to dissipate heat generated by the movement of fluid through the manifold **150**. A coolant source **172** connects to two coolant ports **174** on the manifold **150**. The coolant ports **174**, in turn connect to coolant passages **176** inside the manifold block **151** as shown in FIG. **10**. The coolant passages **176** closely surround the fluid passages **178** connecting to ports A, B, P and T of the main valve **152** to dissipate the heat generated by the fluid flow through the main valve **152**.

[0040] The design of the fluid flow path minimizes the amount of fluid that needs to flow through the main valve **152**, which reduces the overall resistance due to fluid flow. The bypass lines **162A** and **162B** provide a very short fluid flow path between the piston side and rod side of the fluid cylinders **122A** and **122B** for a majority of the fluid flow without having to pass through the manifold assembly **150**. Thus, only a small fluid flow needs to pass through the longer lines **160A** and **160B**. As a result, the frictional losses due to the fluid flow are greatly reduced, which in turn allows greater acceleration in the rate of precession so that the CMG **10** responds more quickly to the roll motion caused by a wave.

[0041] The solenoids **132** on the lockout valve **130** and the solenoid **134** on the main valve **152** provide a mechanism to lock the enclosure **30** to prevent precession of the enclosure **30**. The lockout valves **130** and main valve **152** are biased to a locked position and the solenoid valves **132** and **134** are actuated to unlock the valves **130**, **152**. In the locked position, fluid flow through the

lockout valve **130** and main valve **152** is prevented, which prevents precession of the enclosure **130**. Normally, the enclosure **30** is locked when the flywheel **142** is spinning up to minimize bearing friction. When the flywheel **142** reaches a predetermined speed, the lockout valves **130** and main valve **152** are unlocked. The time it takes to reach the predetermined speed is referred to as the time to engage and is an important consideration for consumer enjoyment. Preferably, a low time to engage is desired so that the benefits of roll reduction are realized sooner. The combination of the increased precession of the enclosure **30** and the faster rate of precession allows more counter torque to be generated more quickly by the CMG **10** as compared to prior art designs. It has been estimated that the peak roll torque can be increased by 5-10% due to the improvements of the braking system **110**.

[0042] FIGS. **11** and **12** illustrate how the improvements of the braking system translate to roll reduction. FIGS. **11** and **12** graphically illustrate the estimated roll resistance for the CMG **10** as herein described with the roll resistance of a similar prior art design for a large rolling wave (FIG. **9**) and a large short wave (FIG. **10**).

Claims

1. A gyroscopic roll stabilizer for a boat, said gyroscopic stabilizer comprising: an enclosure mounted to a gimbal for rotation about a gimbal axis and configured to maintain a below-ambient pressure; a flywheel assembly including a flywheel and flywheel shaft mounted within the enclosure so as to rotate with the enclosure relative to the gimbal axis in response to roll of the boat; one or more bearings for rotatably mounting the flywheel assembly within the enclosure so as to rotate relative to the enclosure about a flywheel axis; a motor for rotating the flywheel assembly about the flywheel axis; a hydraulic cylinder connected between the support frame and the enclosure and configured to resist precession of the flywheel assembly about the gimbal axis when unlocked; a fluid flow path connecting a first port of the hydraulic cylinder to a second port of the hydraulic cylinder enabling flow of hydraulic fluid between the first and second ports as the flywheel assembly precesses; a manifold including a main valve operative to control the resistance of the hydraulic cylinder to precession of the enclosure, the main valve including an input communicatively connected to the first port of the hydraulic cylinder and an output; and a hydraulic accumulator communicatively connected to an output of the main valve configured to compensate for volumetric changes in the hydraulic fluid.
2. The gyroscopic roll stabilizer of claim 1, further including a valve in the fluid flow path enabling fluid flow between the first port and the second port responsive to precession of the flywheel assembly.
3. The gyroscopic roll stabilizer of claim 2, wherein the valve in the fluid flow path is mounted to the hydraulic cylinder so as to move with the hydraulic cylinder as the enclosure rotates about the gimbal axis.
4. A method of operating a controlled moment gyroscope configured for roll stabilization of a boat, the controlled moment gyroscope comprising a flywheel assembly mounted within an enclosure for rotation with the enclosure about a gimbal axis, said method comprising: controlling precession of the flywheel assembly using a hydraulic cylinder connected between the enclosure and a support frame; controlling the resistance of the hydraulic cylinder to precession of the enclosure by controlling the flow of a hydraulic fluid through a main valve of a manifold communicatively connected to a first port of a hydraulic cylinder; directing fluid flow along a fluid flow path between the first port of the hydraulic cylinder and a second port of the hydraulic cylinder responsive to precession of the flywheel assembly; and compensating for volumetric changes in the hydraulic fluid by collecting hydraulic fluid in a hydraulic accumulator disposed on a low pressure side of the manifold and communicatively connected to an output of the main valve.
5. The method of claim 4 further comprising controlling the flow of hydraulic fluid along the fluid

flow path using a valve disposed in the fluid flow path.

6. The method of claim 5, further comprising moving the valve with the hydraulic cylinder as the enclosure rotates about the gimbal axis.

7. A gyroscopic roll stabilizer for a boat, said gyroscopic stabilizer comprising: an enclosure mounted to a gimbal for rotation about a gimbal axis and configured to maintain a below-ambient pressure; a flywheel assembly including a flywheel and flywheel shaft mounted within the enclosure so as to rotate with the enclosure relative to the gimbal axis in response to roll of the boat; one or more bearings for rotatably mounting the flywheel assembly within the enclosure so as to rotate relative to the enclosure about a flywheel axis; a motor for rotating the flywheel assembly about the flywheel axis; a hydraulic cylinder connected between the support frame and the enclosure and configured to resist precession of the flywheel assembly about the gimbal axis when unlocked; and a manifold assembly including: a manifold block having a main valve operative to control the resistance of the hydraulic cylinder to precession of the enclosure, the main valve including an input communicatively connected to the first port of the hydraulic cylinder and an output; a hydraulic accumulator communicatively connected to an output of the main valve and configured to compensate for volumetric changes in the hydraulic fluid; and a coolant passage in the manifold block for circulating a cooling fluid through the manifold block to dissipate heat from the manifold block.

8. The gyroscopic roll stabilizer of claim 7, further comprising a fluid flow path connecting a first port of the hydraulic cylinder to a second port of the hydraulic cylinder enabling flow of hydraulic fluid between the first and second ports as the flywheel assembly precesses.

9. The gyroscopic roll stabilizer of claim 8, further comprising a valve in the fluid flow path enabling fluid flow between the first port and the second port responsive to precession of the flywheel assembly.

10. The gyroscopic roll stabilizer of claim 9, wherein the valve in the fluid flow path is mounted to the hydraulic cylinder so as to move with the hydraulic cylinder as the enclosure rotates about the gimbal axis.

11. A method of operating a controlled moment gyroscope configured for roll stabilization of a boat, the controlled moment gyroscope comprising a flywheel assembly mounted within an enclosure for rotation with the enclosure about a gimbal axis, said method comprising: controlling precession of the flywheel assembly using a hydraulic cylinder connected between the enclosure and a support frame; controlling the resistance of the hydraulic cylinder to precession of the enclosure by controlling the flow of a hydraulic fluid through a main valve of a manifold communicatively connected to a first port of a hydraulic cylinder; compensating for volumetric changes in the hydraulic fluid by collecting hydraulic fluid in a hydraulic accumulator disposed on a low pressure side of the manifold and communicatively connected to an output of the main valve; and, dissipating heat from the manifold by circulating a cooling fluid through the manifold.

12. The method of claim 11, further comprising directing fluid flow along a fluid flow path between the first port of the hydraulic cylinder and a second port of the hydraulic cylinder responsive to precession of the flywheel assembly.

13. The method of claim 12 further comprising controlling the flow of hydraulic fluid along the fluid flow path using a valve disposed in the fluid flow path.

14. The method of claim 13, further comprising moving the valve with the hydraulic cylinder as the enclosure rotates about the gimbal axis.
